Midterm Exam Study Guide

Econ 6343: Econometrics III, Fall 2026

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When: Tuesday, November 10, 2026, in class (75 minutes). **Format:** 7 questions, 70 points. Closed AI, notes, book, and device; a basic scientific calculator is allowed. **Coverage:** everything through Class 22 (Learning models).

Most questions ask you to *explain*, *critique a claim*, or decide whether a statement is *true or false* and say why. Expect one short derivation and one question that asks you to read a short Julia function. Good answers are a few clear sentences, not long essays.

1. Stated preferences and stated probabilities

- Why researchers collect stated-preference data, and how they differ from revealed-preference data
- What a stated probability measures, and what it adds beyond a discrete choice or a ranking
- How logit-type models are estimated when the outcome is a probability rather than a 0/1 choice, and the practical issues that come up

Review: Lecture 11; Stated Preference videos; *Train* 7.2–7.3; mobility (PCM) in-class activity.

2. Identification in dynamic discrete choice models

- What it means for a parameter of a dynamic model to be identified, and which parameters are hardest to pin down
- The role of functional-form assumptions, and what kinds of variation in the data help

Review: Lecture 7; Dynamic Discrete Choice videos (the whole series); Rust (1987).

3. Experiments, structural models, and counterfactuals

- Internal vs. external validity; what experiments can and cannot tell us
- How experiments and structural models can complement each other (estimation, validation, prediction)
- What must be true for a counterfactual simulation to be credible

Review: Lectures 2, 3, and 10; Model Fit, Validation, and Counterfactuals videos; counterfactuals in-class activity.

4. Breadth across the course (short true/false items)

- Interpreting the key parameters of nested and mixed logit models, and the restrictions theory places on them
- Measuring goodness of fit in discrete choice models; consumer surplus and the logsum
- Properties of simulation-based estimators (SML, SMM, indirect inference) and how they depend on the number of draws
- What identifies preference heterogeneity; finite mixtures and the EM algorithm
- Solving finite- vs. infinite-horizon dynamic models

Review: Lectures 4–7 and 9; the corresponding videos; *Train* Ch. 3, 4, 6, and 10.

5. Reading and testing code

- Read a short Julia function like those in the problem sets and in-class demos, and explain what each step computes
- Connect the math of a model (probabilities, likelihoods, normalizations) to the lines of code that implement it
- Spot mistakes that do *not* produce an error message, and design unit tests that would catch them

Review: PS2–PS4 and PS7 (including the demo files); Clean Code lecture.

6. Conditional choice probabilities

- The Hotz–Miller inversion: what it says and why it matters computationally
- Terminal, renewal, and finite-dependence actions
- What CCP methods do and do not simplify (estimation vs. counterfactuals)

Review: Lecture 8; CCP videos; Hotz and Miller (1993); Arcidiacono and Miller (2011); PS6 and its demo.

7. Unobserved heterogeneity: factor and learning models

- Proxy variables, measurement error, and factor analysis (including normalizations and identification)
- Learning models and Bayesian updating, and how beliefs evolve over time
- How factor models and learning models differ, and how they relate to each other

Review: Lectures 12–13; Measurement Error and Learning Models videos; Heckman, Stixrud and Urzua (2006); Miller (1984); PS8.

8. Facts worth knowing cold

- Standard Type 1 Extreme Value: mean $\gamma \approx 0.5772$ (Euler's constant), variance $\pi^2/6$; the difference of two is logistic
- Logit choice probabilities $P_{ij} = \exp(v_{ij}) / \sum_k \exp(v_{ik})$; only utility differences matter, and the scale is normalized
- With i.i.d. T1EV errors, $\mathbb{E} \max_j \{v_j + \epsilon_j\} = \ln \sum_k \exp(v_k) + \gamma$

9. Not on the exam

Material from Class 24 onward: referee reports, the final-project demonstration, advanced optimization, extreme value theory beyond Type 1 EV, relative correlation restrictions, and machine learning. Presentation and referee-report content is also excluded.